

Laboratory Work 3

Creating Custom Decorators

Objective: to gain basic knowledge and skills in creating decorators.

To determine your variant number, execute the following statement:

$$N = \text{ord}(\text{"XXX"}) \% 3 + 1,$$

where XXX is the first capital letter of your name (in Latin script). For example, for James Bond, the variant number would be $N = 3$

Variant 1

Create a custom decorator to log function details (name, result, call time, parameters, and execution status) to a file. Test it on 3–4 functions with varying numbers and types of parameters.

Variant 2

Create a custom decorator that limits function execution to a maximum of 3 times. Test it on 3–4 functions with varying numbers and types of parameters.

Variant 3

Create a custom decorator for function result caching: if a specific function has already been executed with certain parameters, do not execute the function again; instead, return the result from the cache. Test it on 3–4 functions with varying numbers and types of parameters.

Report Contents

1. Title page.
2. Objective of the work.
4. Code and screenshots of script execution.
5. Conclusions.

Note: Instead of a formal report, you may complete the assignment in a Jupyter Notebook and push it to GitHub.